

CLASSIFYING DECELERATION MEASUREMENT SEQUENCES

Team **HalloLeute**

Nikita Biswas · Sebastian Garcia · Chaeran Yoo · Daniel Useche





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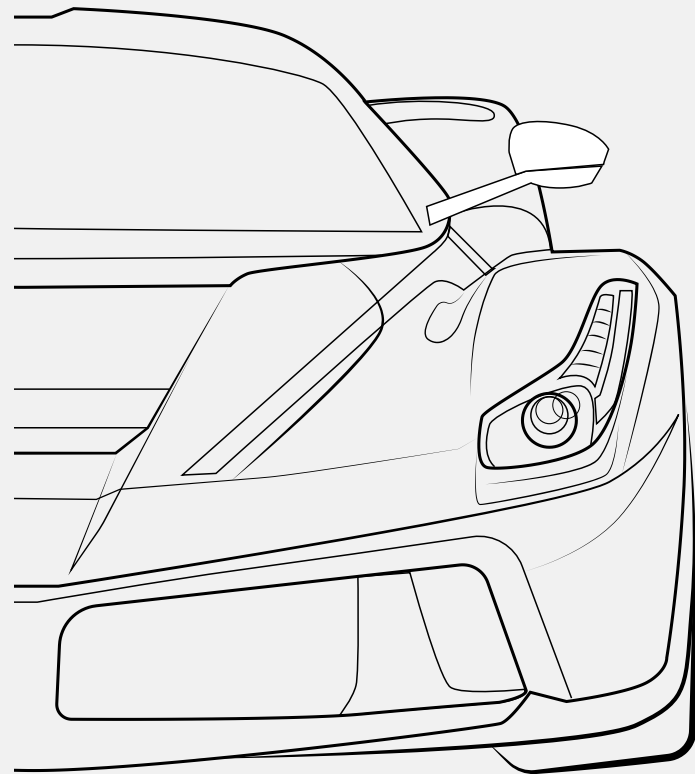
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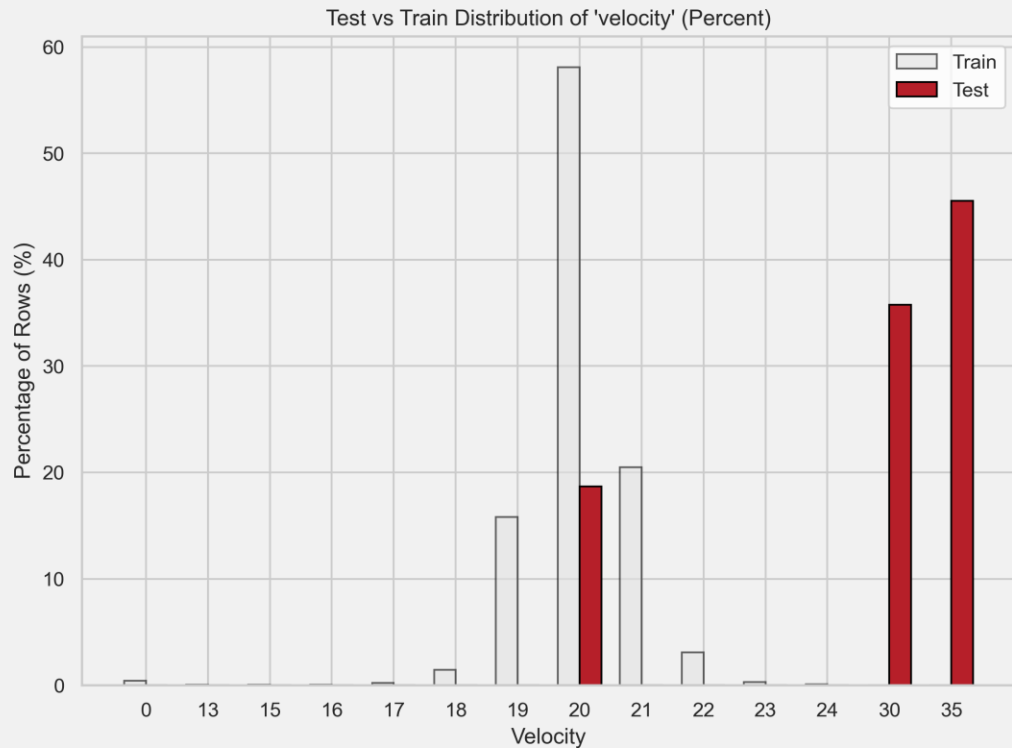
01

EXPLORATORY **DATA ANALYSIS**



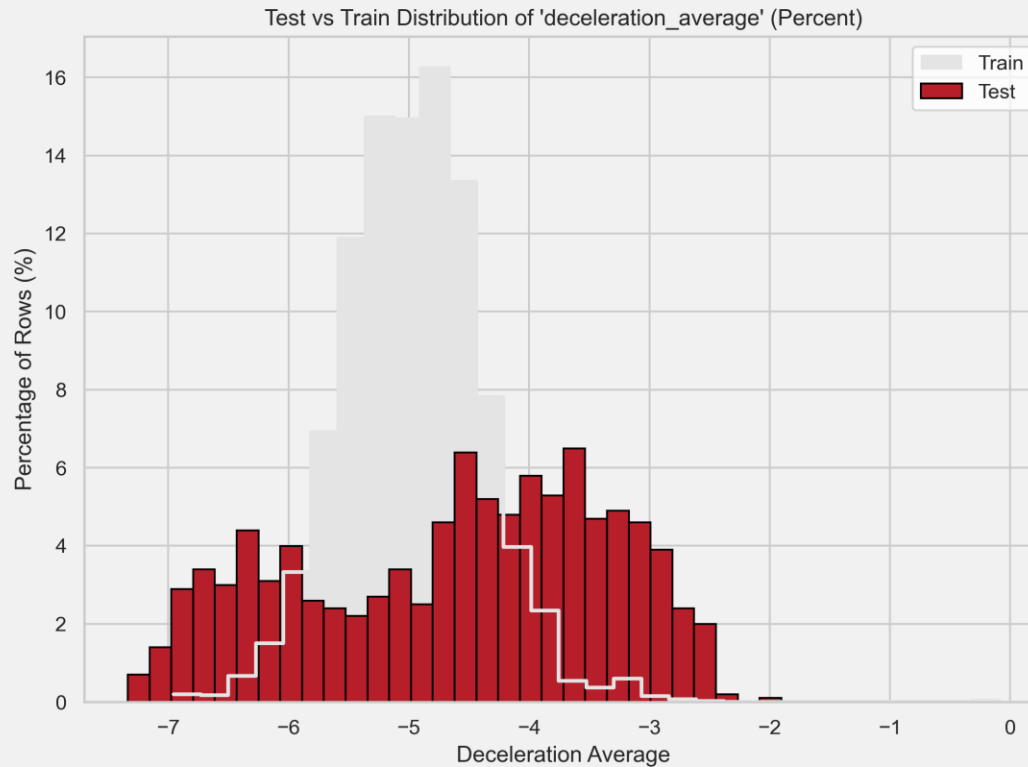


VELOCITY **ANALYSIS**





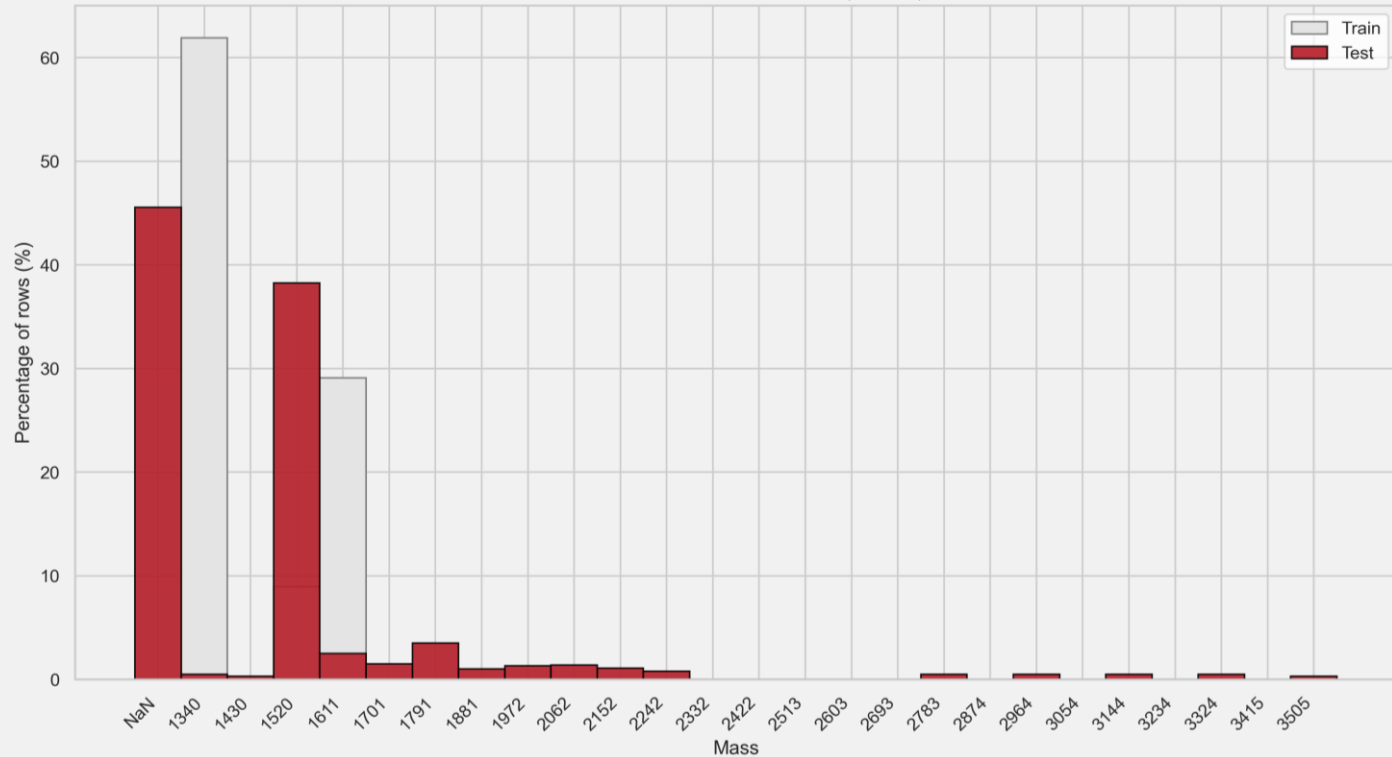
DECELERATION AVERAGE **ANALYSIS**





MASS ANALYSIS

Test vs Train Distribution of 'mass' (Percent)



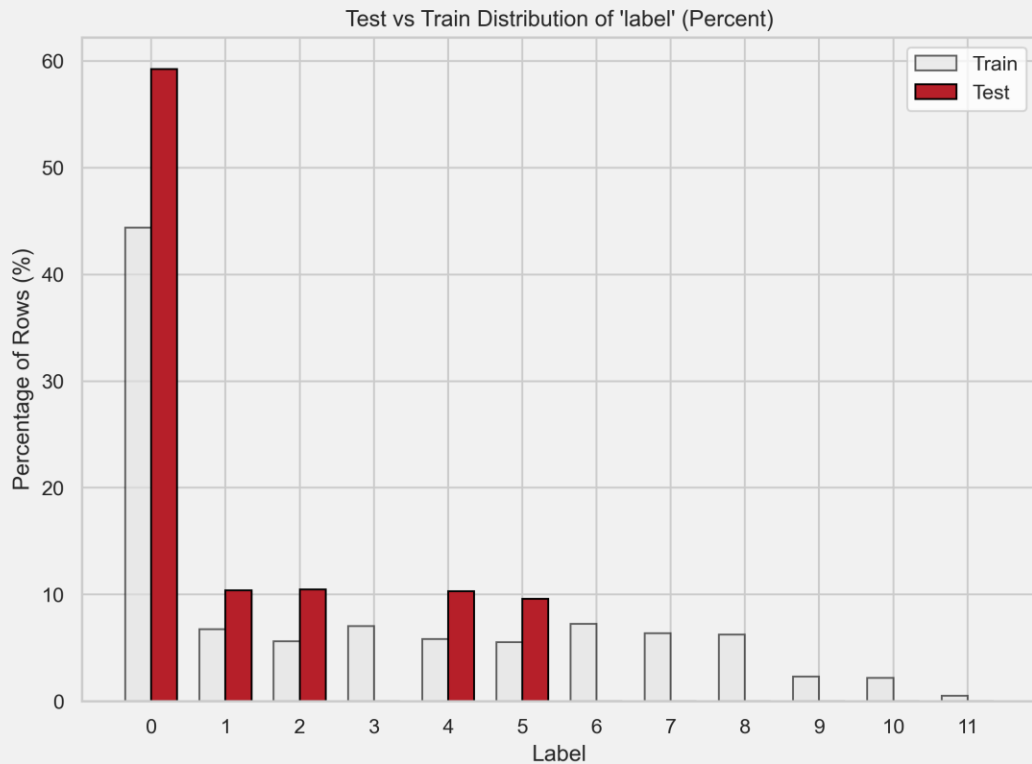
45.6%

NaN

All of them
fourfour
model



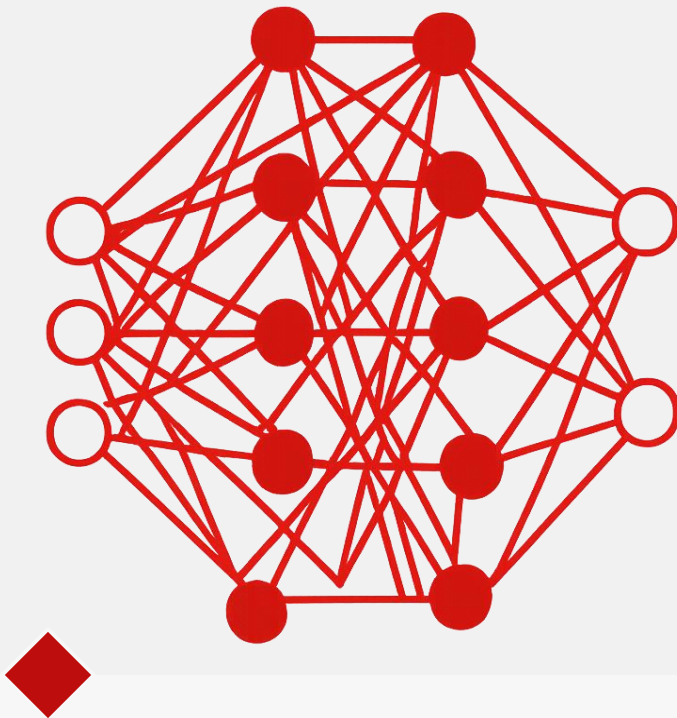
LABEL ANALYSIS



02

BASE **MODELS**

FCN, CNN, LSTM



BEST **MODEL**



F1 SCORE

Harmonic mean of precision and recall,
especially useful when dealing with
imbalanced dataset.



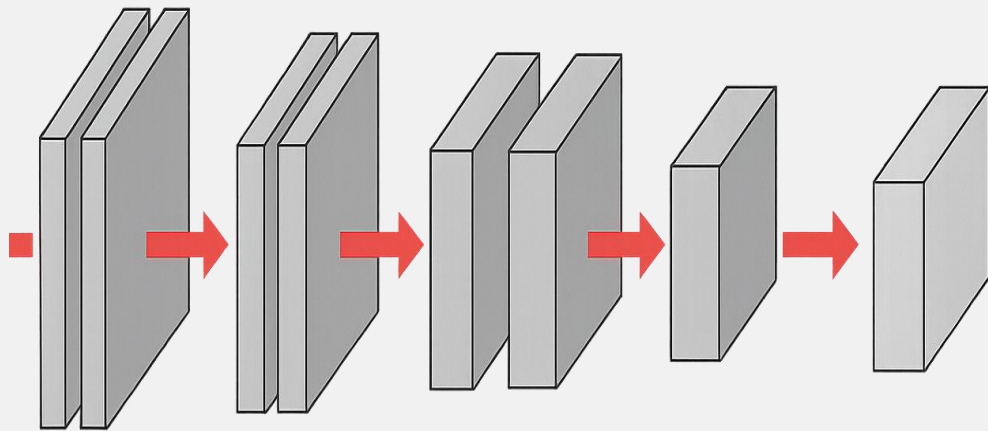
TEST ACCURACY

Reflects how well the model generalizes to
new, unseen data, crucial for evaluating its
real-world usability.

FCN

ARCHITECTURE

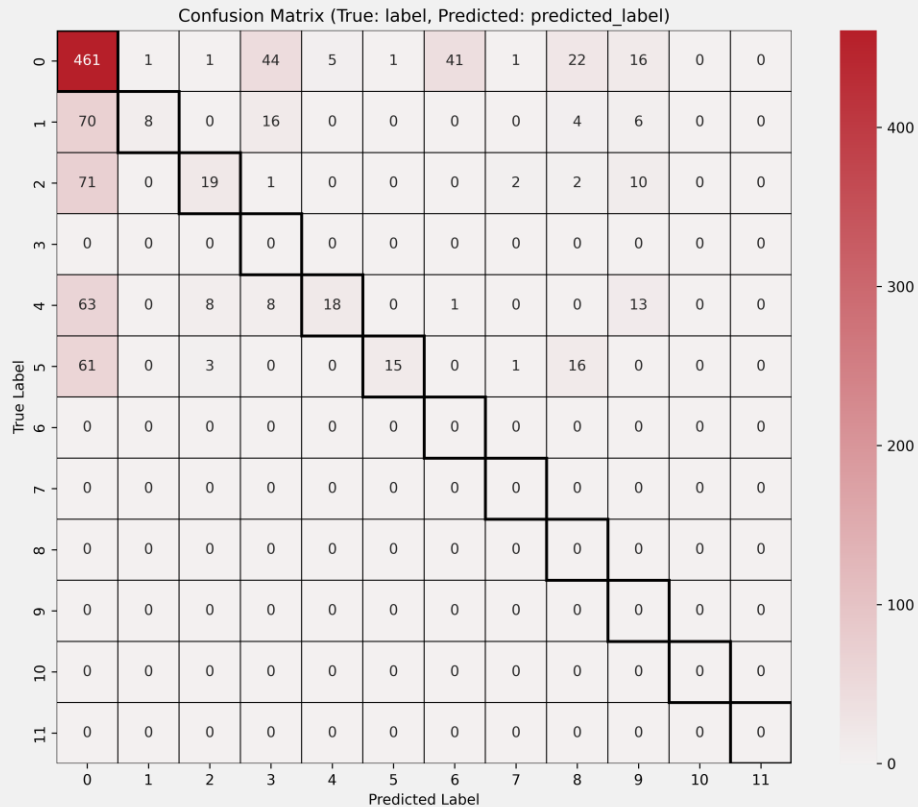
- Input layer.
- Three sequential 1D **convolutional** blocks.
- Global average **pooling** layer.
- Final fully connected (**linear**) output layer



RESULTS

Hyperparameter	Value
Batch size	32
Learning rate	10^{-4}
Number of epochs	20
Optimizer	Adam
Activation function	ReLU
Kernel size	8, 5, and 3
Output channels (Conv1)	64
Output channels (Conv2)	128
Output channels (Conv3)	128

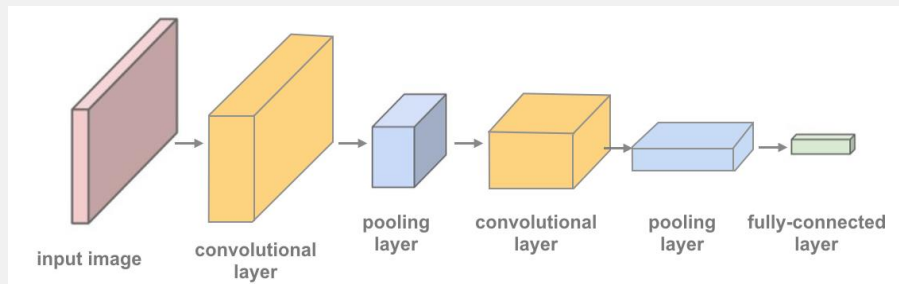
Metrics	Value
Average Test Accuracy	0.4840
Average macro-F1 Score	0.3401
Average Validation Accuracy	0.9911



2D-CNN

ARCHITECTURE

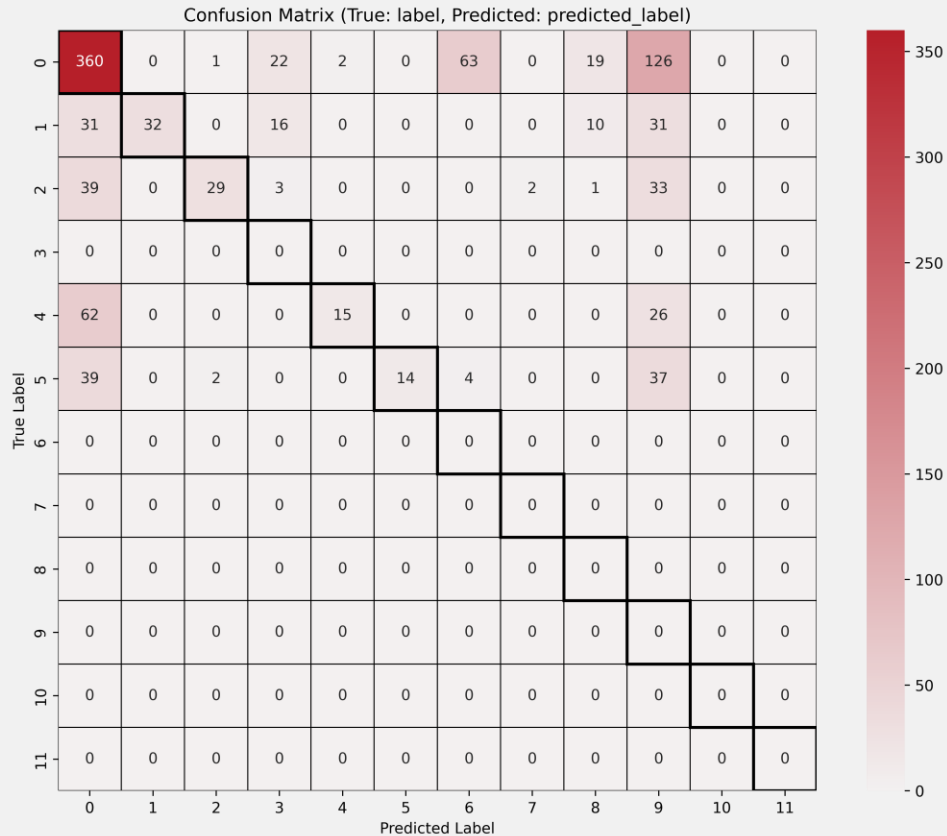
- Learn how sensors change over time **and** how they relate to each other at the same time
- **Conv2D (1×6)** → sensor correlation
- **Conv2D (3×1)** → temporal pattern
- Flatten → FC → SoftMax (12 classes)
- Conv2D followed by batch normalization, ReLU, pooling, and dropout.



RESULTS

Hyperparameter	Value
Batch size	64
Learning rate	10^{-3}
Number of epochs	100
Output channels (Conv1)	32
Output channels (Conv2)	64
FC hidden units	64
Dropout rate	0.01

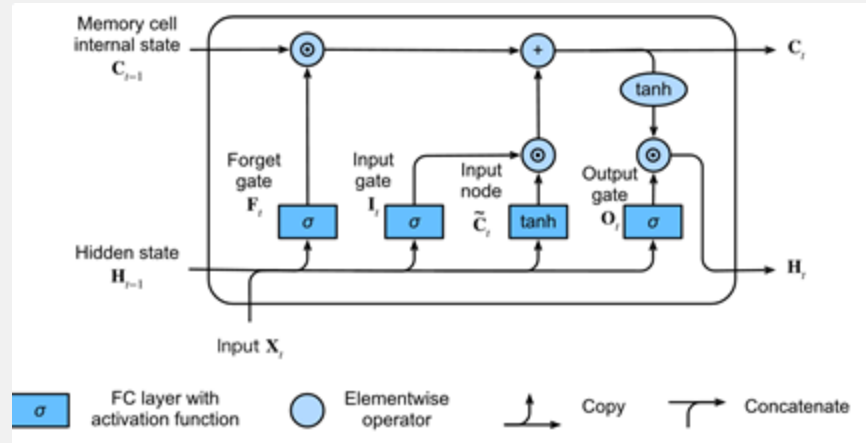
Metrics	Value
Average Test Accuracy	0.4390
Average macro-F1 Score	0.3898
Average Validation Accuracy	0.9975



LSTM

ARCHITECTURE

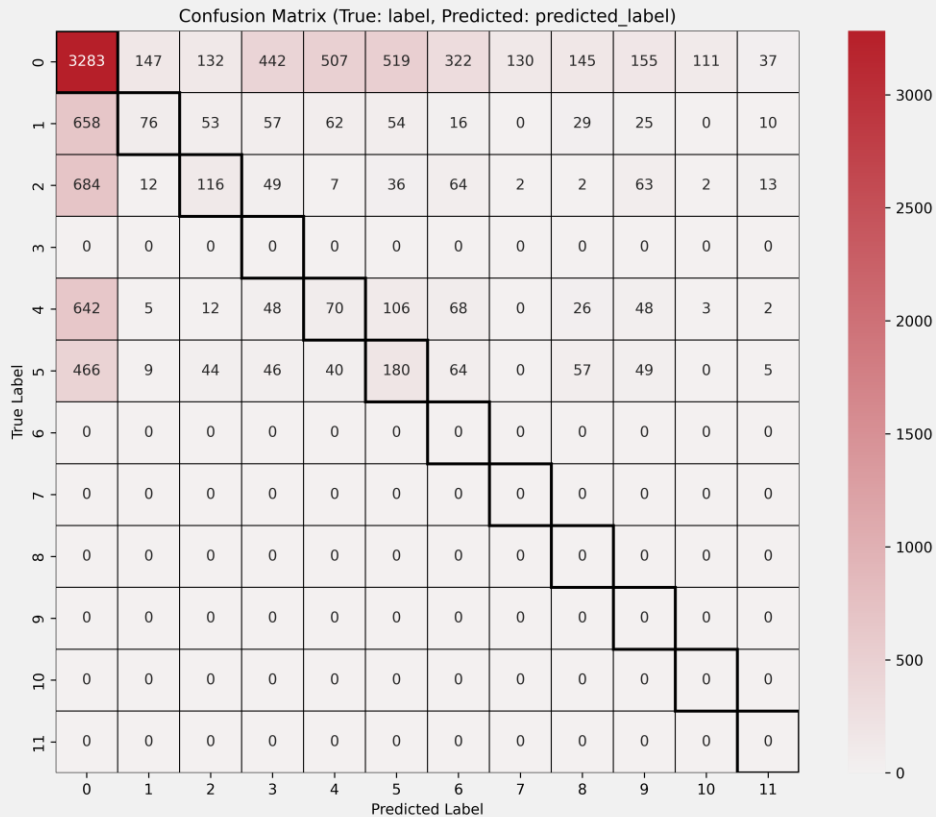
- Input: 128 timesteps $\times$ 6 sensor channels
- Gate-controlled memory (input, forget, output gates)
- Single fully-connected layer: maps summary to brake combination



RESULTS

Hyperparameter	Value
Batch size	64
Learning rate	10^{-3}
Number of epochs	40
Optimizer	Adam
Number of layers	4

Metrics	Value
Average Test Accuracy	0.3721
Average macro-F1 Score	0.2165
Average Validation Accuracy	0.9973

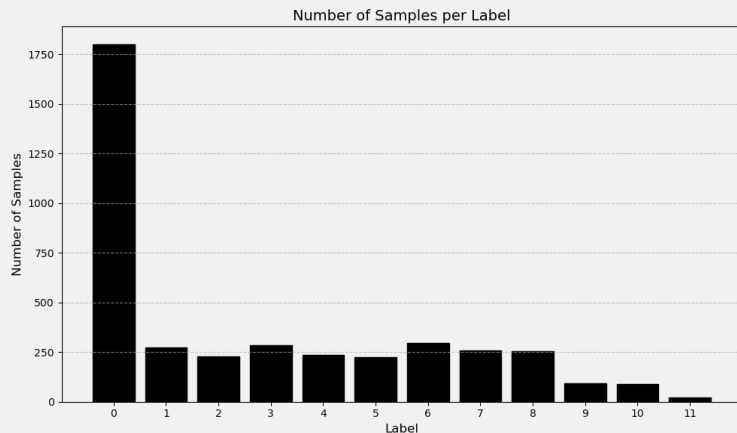


03

BIAS IN THE **DATA SET**

BIAS reduces generalization

Class imbalance (class 0).
Correlation with Metadata.
Train/Test Distribution Shift.





PREPROCESSING & DATA **AUGMENTATION**



Standardization/Min-Max Normalization

No significant improvement in the overall performance



Time shifting

Tests model sensitivity to minor misalignments in measurements. Slight boost test accuracy by small shifts but larger shifts impacted F1-score.



Gaussian Noise

Adds zero-mean Gaussian noise to every sensor channel. Large noise degraded the performance.



Scaling

Varies signal amplitude and simulates differences in sensor calibration or brake intensity. Tight scaling on az, ry, rz yielded better F1 score, whereas extreme scaling hurt accuracy.



DATA **AUGMENTATION**



Truncation

Removes a fixed fraction (e.g. 10-30 %) of the beginning of each signal. Even small truncation eroded performance



Sliding window

Splits each signal into smaller windows, treats each window as an independent training example. Drastic reduction in Macro F1-score.



Selective augmentation

Selected minority classes were over-sampled, moderate noise was introduced, down-sampling of majority class. Limited improvement in misclassification.

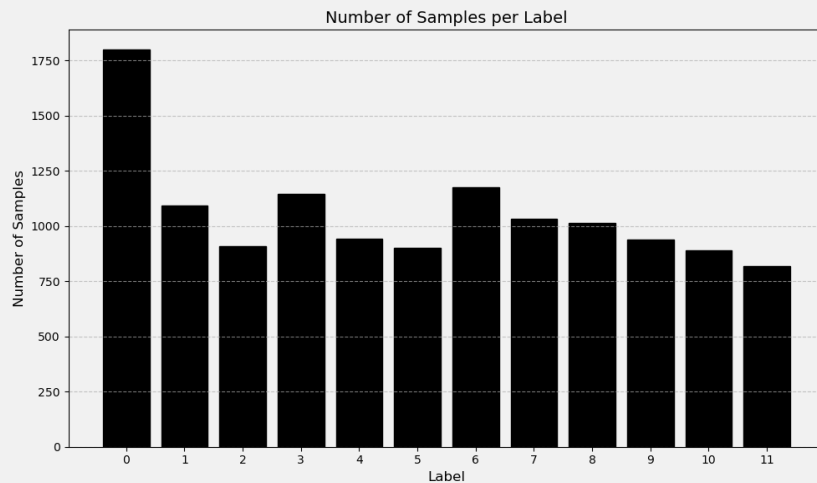
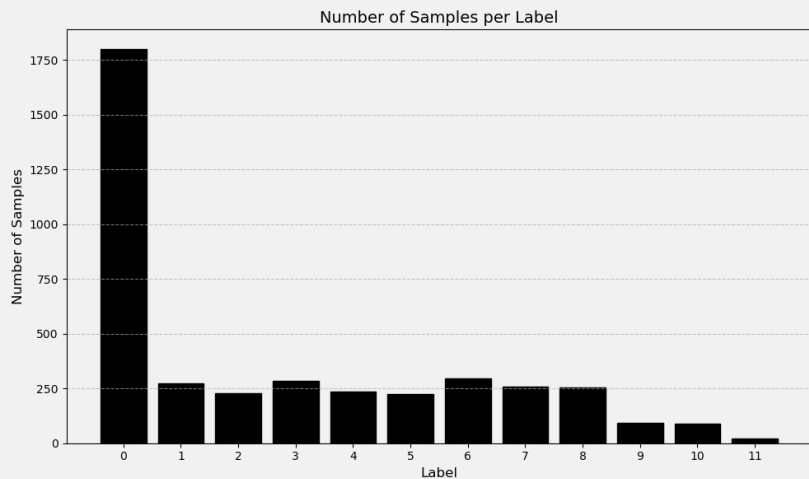


DATA **AUGMENTATION** - RESULTS

Augmentation	Augmentation parameter	Test accuracy	Macro F1 score
Gaussian Noise	0.001	0.4693	0.3134
	0.005	0.4739	0.3139
	0.01	0.4939	0.3067
	0.02	0.4627	0.3128
Time shifting	2	0.4702	0.3149
	5	0.477	0.2973
	10	0.4654	0.3015
Scaling (axes 3,4)	(0.8,1.2)	0.4757	0.3231
Scaling (axes 3,4)	(0.5,2.0)	0.4578	0.2944
Scaling (axes 2,4,5)	(0.5,2.0)	0.4563	0.3256
Truncation	10%	0.4601	0.2959
	20%	0.4205	0.2546
Sliding window (Window size x Stride)	32 x 16	0.4921	0.2496
	64 x 32	0.4521	0.2515
Selective augmentation (Minority oversampling + moderate noise)		0.4851	0.3084



Selective augmentation with different techniques



In this experiment, an **average test accuracy** over 10 random splits of 0.4301 was obtained with an **average macro F1 score** of 0.1684.

04

Debiasing

**Variational
Autoencoder(VAE)**

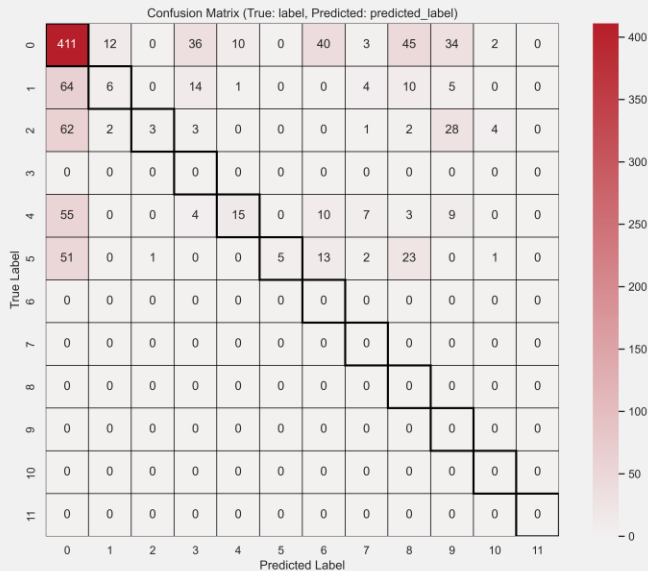


Debiasing VAE

ARCHITECTURE

- ♦ **FCN based**
- ♦ **Encoder:** 3 Conv1D layers followed by batch normalization and ReLU (FCN backbone)
 - Final output: latent mean (μ) & log-variance ($\log\sigma^2$)
- ♦ **Decoder:** Linear layer to reconstruct the original input (128×6) from latent vector
- ♦ **Classifier:** Fully connected layer
 - Latent vector $\rightarrow$ 12 logits
 - Train together with the encoder and decoder
- ♦ **Loss function:** $L_{\text{total}} = \text{MSE} + \beta \cdot \text{KL} + \lambda \cdot \text{CE}$

RESULTS



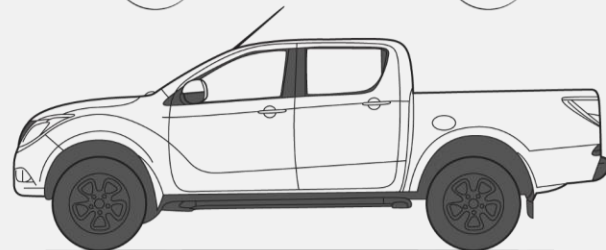
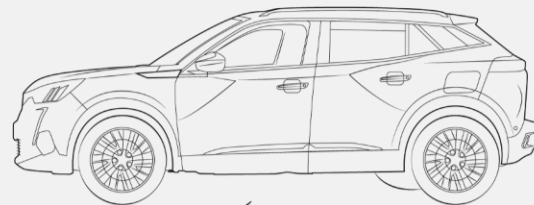
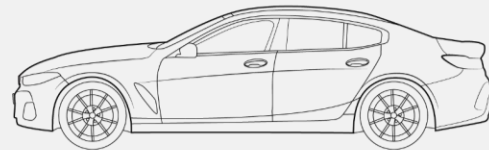
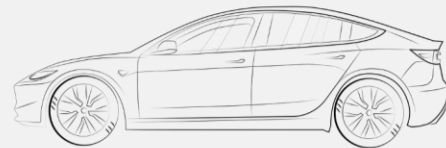
- VAE learns **slightly better**, but still biased to majority classes
- **Batch size (32), LR (0.0001)** → **higher** accuracy, but **lower** macro F1
- **Adaptive Sampling** (latent-based) → Focused on underrepresented regions
- **Class-weighted Loss** → Emphasized rare classes
- Both methods underperformed compared to **uniform sampling**

Hyperparameter	Value
Batch size	64
Number of epochs	100
Learning rate	10^{-3}
Latent dimension	16
KL weight (β)	10^{-3}
Classification weight (λ)	1.0

Model	Test Accuracy	Macro F1
FCN (Baseline)	0.4840	0.3401
FCN VAE	0.4861	0.3544
Smaller LR & Batch	0.5063	0.2612(↓)
Adaptive Sampling	0.4496	0.2947
Class-weighted Loss	0.4613	0.3377

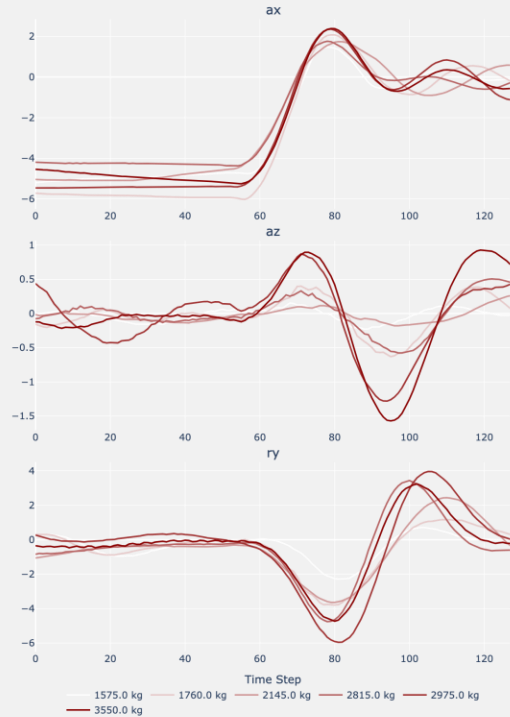
05

BINARY - PHYSICAL **MODELS**

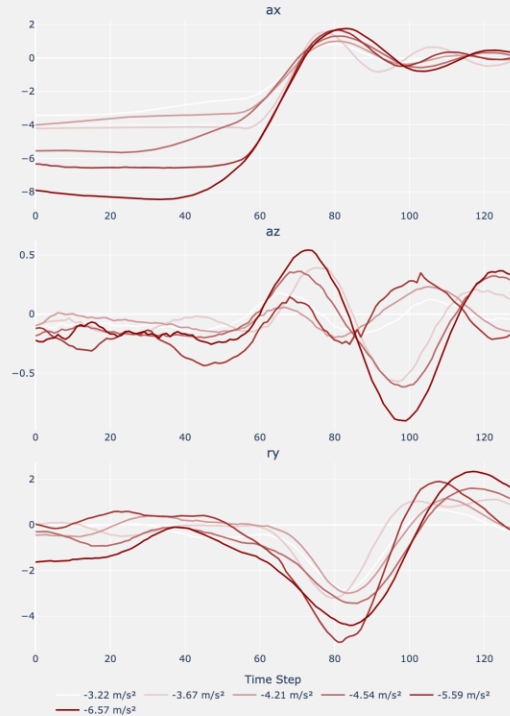


PHYSICAL FEATURES **MODEL**

Effect of Mass on Signals



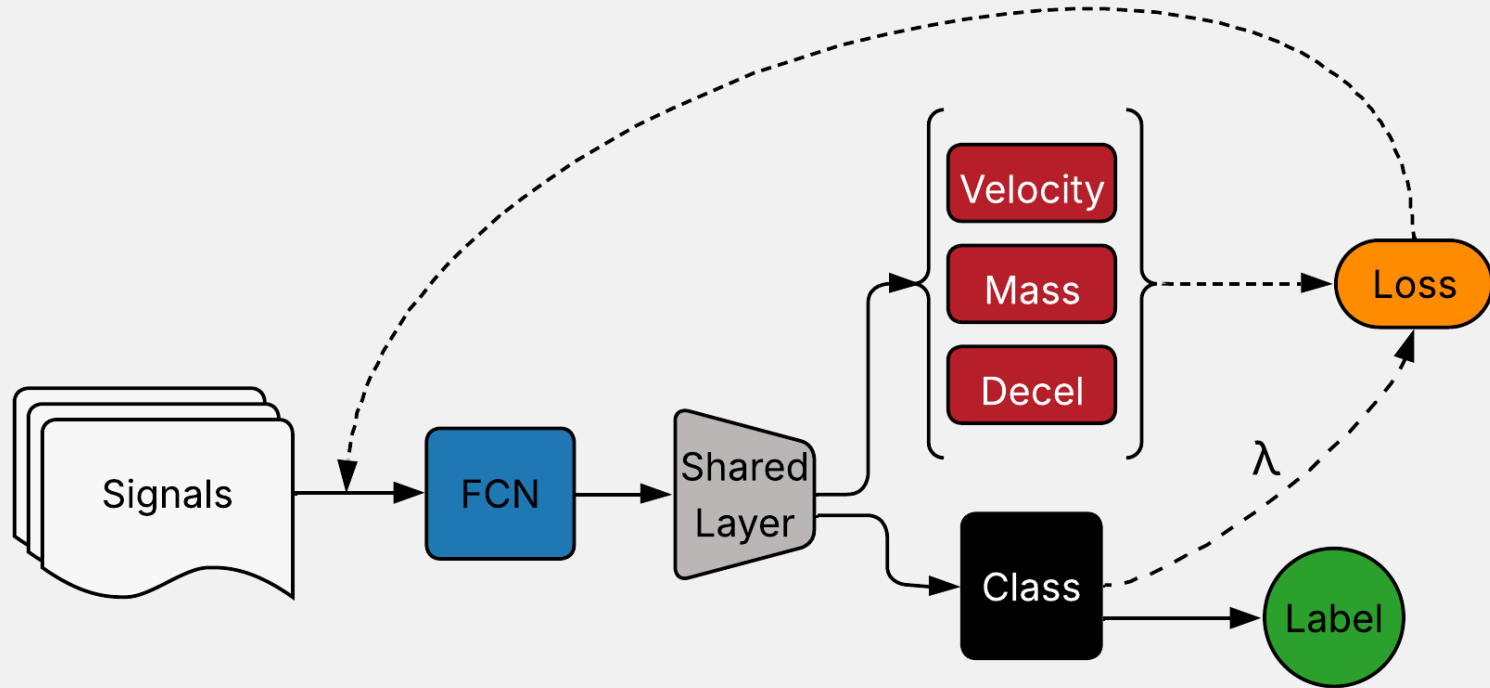
Effect of Deceleration Average on Signals



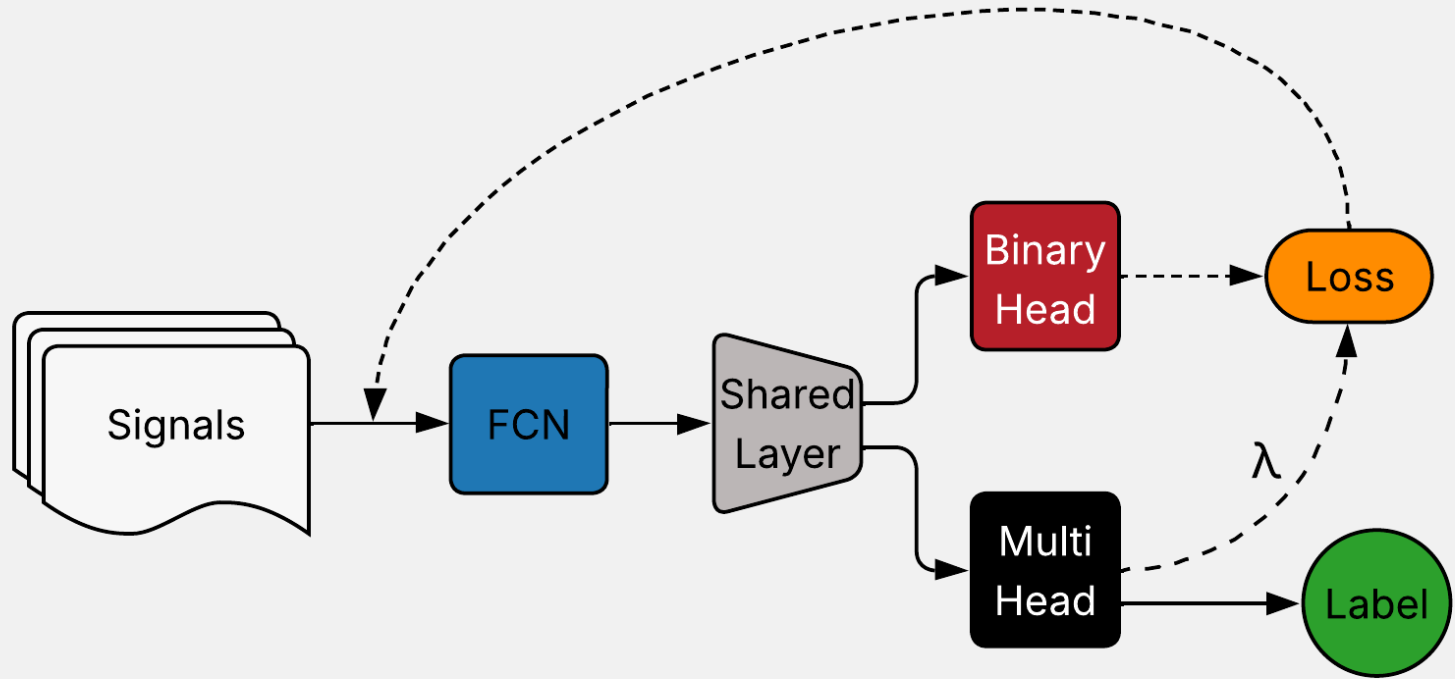
Effect of Velocity on Signals

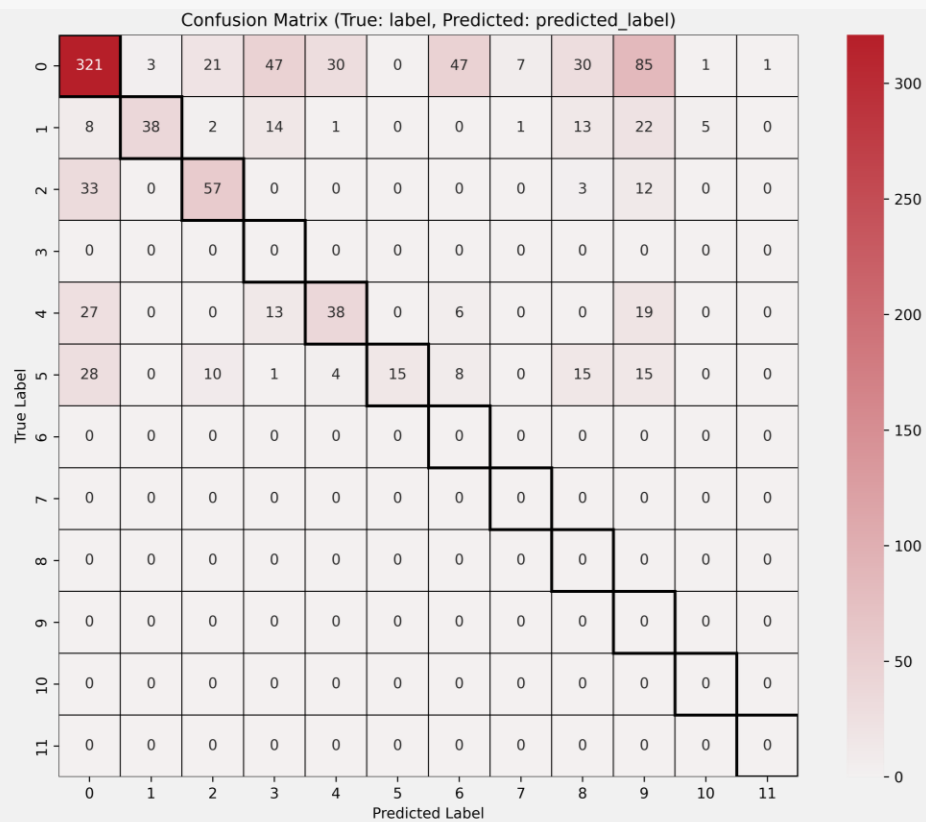


PHYSICAL FEATURES **MODEL**



BINARY MODEL





Hyperparameter	Value
Learning rate	10^{-3}
Number of epochs	5
λ (loss weight)	1
Oversampling target	labels = 0
Oversample multiplier	1
Standardize	True
Scaling range	(0.8, 1.8)
Scaled channels	ax, az, ry

Metrics	Value
Iterations	20
Average Test Accuracy	0.4319
Average Test macro-F1 Score	0.4047

CONCLUSIONS



CLASS IMBALANCE



LACK OF
GENERALIZATION



MODEST GAINS FROM:
Augmentation
VAE
Additional architectures



FUTURE **WORK**



METADATA INCLUSION



MORE DIVERSE
TRAIN DATASET



MORE SOPHISTICATED
MODELS



A close-up photograph of a person's hands on a car's steering wheel. The person is wearing a dark suit jacket, a white shirt, and a gold watch. The background is slightly blurred, showing the interior of the car. A large white rectangular box is overlaid on the center of the image, containing the text 'THANK YOU!'. The word 'THANK' is in black, and 'YOU!' is in red. A small red diamond shape is located in the top right corner of the white box.

THANK
YOU!
